

Steps

Design Req.

↓

Op. Points

↓

Magnetic. Builder

↓

Summary

Tool menu

Magnetic Summary



My custom magnetic

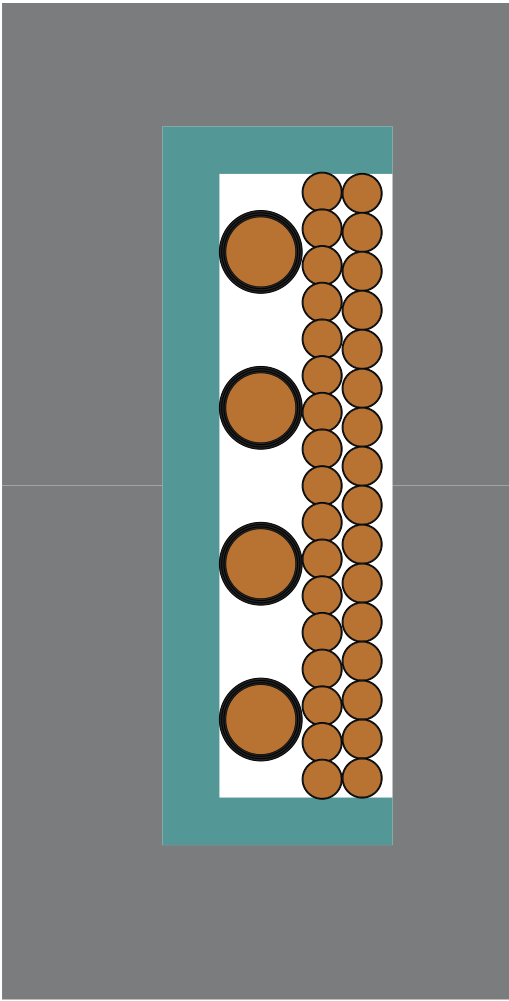
ETD 34/17/11 N87

Power Transformer

OpenMagnetics

Generated: January 29, 2026

CORE LOSS	WINDING LOSS	PEAK B	CORE
11.11 W	11.47 W	0.24 T	ETD 34/17/11



Show H Field

Show E Field

Electrical Specifications

Parameter	Symbol	Typical	Unit	Conditions
Operating Frequency	f	100	kHz	
Peak Flux Density	B_{pk}	0.24	T	At operating point

Primary $f = 100.0 \text{ kHz}$

⚡ Voltage

Sinusoidal

V_{PK-PK}	V_{PK}	V_{RMS}	V_{OFFSET}	DUTY
98.27 V	49.14 V	45.72 V	1.05 mV	50.0%



⚡ Current

Custom

I_{PK-PK}	I_{PK}	I_{RMS}	I_{OFFSET}	DUTY
125.37 A	62.69 A	54.22 A	74.35 μA	94.0%

Secondary $f = 100.0 \text{ kHz}$

⚡ Voltage

Sinusoidal

V_{PK-PK}	V_{PK}	V_{RMS}	V_{OFFSET}	DUTY
818.9 V	409.46 V	404.16 V	-6.66 mV	50.0%



⚡ Current

Custom

I_{PK-PK}	I_{PK}	I_{RMS}	I_{OFFSET}	DUTY
15.06 A	7.53 A	6.5 A	367.54 nA	94.0%



Core Material	N87
Number of Stacks	1
Effective Length (le)	80.07 mm
Effective Area (Ae)	97.26 mm ²
Effective Volume (Ve)	7787.64 mm ³
Minimum Area	91.6 mm ²
Total Gap Length	0 m
Number of Gaps	0

Core Gapping

#	Type	Length
1	Residual	5 µm
2	Residual	5 µm
3	Residual	5 µm

Material Properties

Manufacturer	TDK
Curie Temperature	210 °C
Density	4.85 Mg/m ³

Winding Configuration

Winding	Turns	Parallels	Wire
Primary	4	1	Litz Round 2.00 - Grade 1
Secondary	33	1	Litz 75xØ0.10 mm

Performance Analysis

Test Conditions: $f = 100.0\text{ kHz}$, $T_{amb} = 100^{\circ}\text{C}$

Core Losses (Pcore)	11.11 W
Ohmic Losses	11.065 W
Skin Effect Losses	307.309 µW
Proximity Losses	0.401 W
Total Losses	22.576 W

Winding Performance

Winding	DCR	Loss	Leakage
Primary	2.48 mΩ	7.3 W	-
Secondary	89.36 mΩ	4.16 W	-

- 3

 Wind Secondary: 17 turns
- 5

 Wind Secondary: 16 turns
- 4

 Apply Insulation tape
- 6

 Apply Insulation tape

Note: This datasheet was automatically generated by OpenMagnetics based on the design specifications and simulation results. Actual performance may vary. Please verify all specifications before production use.